Assignment #1

Question 1

Choose the right answer:

1. Which of the following is a Computer Graphics type?
   a) Raster and Vector
   b) Raster and Scalar
   c) Scalar only
   d) All of the abov
2. Which of the following is defined as the process of elimination of parts of a scene outside a window or a viewport?
   a) editing
   b) cutting
   c) plucking
   d) clipping
3. In a graphical system, an array of pixels in the picture are stored in which of the following locations?
   a) Frame buffer
   b) Processor
   c) Memory
   d) All of the mentioned
4. Bitmap is a collection of _____ that describes an image.
   a) pixels
   b) algorithms
   c) lines
   d) colors
5. Which of the following devices provides positional information to the graphics system?
   a) Pointing devices
   b) Both Input devices and Pointing devices
   c) Output devices
   d) Input devices
6. Viewing transformation is the process of mapping a Viewport to world window.
   a) False
   b) True
7. Which of the following is a primary output device of a graphics system?
   a) Printer
   b) Scanner
   c) Video monitor
   d) Neither Scanner nor Video monitor

8. Which of the following is commonly known as frame buffer on a black and white system with one bit per pixel?
   a) Bitmap
   b) Pix map
   c) Multi map
   d) All of the mentioned
9. Which of the following algorithm is a uses floating point arithmatic for drawing line?
   a) Parallel line algorithm
   b) Mid-point algorithm
   c) DDA line algorithm
   d) Bresenham's line algorithm
10. Which of the following algorithm is a more accurate algorithm for drawing line?
   a) Parallel line algorithm
   b) Mid-point algorithm
   c) DDA line algorithm
   d) Bresenham's line algorithm

Question 2

- Using Bresenham's line drawing algorithm, find the intermediate points between (28,3) and (22,7)
- Using DDA line drawing algorithm, find the intermediate points between (6,9) and (0,2)
- Using the mid-point circle drawing algorithm, draw a circle with a radius of 4 units, along the circle octant in the 1st quadrant from x = 0 to x = y.
- Determine pixel positions along the ellipse path in the first quadrant using midpoint ellipse algorithm.
    - Given:
        - input ellipse parameter r_x=10 and r_y=8
        - $p1_0 = r_y^2 - r_x^2 r_y + \frac{1}{4} r_x^2$
        - $p1_0 < 0 \qquad p1_{k+1} = p1_k + 2r_y^2 x_{k+1} + r_y^2$
        - $p1_0 > 0 \qquad p1_{k+1} = p1_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_y^2$
        - $p2_0 = r_y^2 \left(x_0 + \frac{1}{2}\right)^2 + r_x^2 (y_0 - 1)^2 - r_x^2 r_y^2$
        - $p2_0 < 0 \qquad p2_{k+1} = p2_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_x^2$
        - $p2_0 > 0 \qquad p2_{k+1} = p2_k - 2r_x^2 y_{k+1} + r_x^2$

Question 3

Write a program for drawing the following figure attach a shot of the code and the resulting figure

